# WP5.3 — Deliverables with Sub-Tasks

---

# D1 — Data Ingestion & Dataset Preparation

## Objective

Prepare modeling-ready datasets from WP5.1 or sample setups.

## Sub-Tasks

- Define required process signals and KPIs
- Identify input/output variables for SI
- Define data schema and naming conventions
- Develop data ingestion pipeline/interface
- Synchronize timestamps across signals
- Handle missing/noisy data
- Data filtering and preprocessing
- Feature extraction and normalization
- Dataset labeling (defects/events)
- Generate train/validation/test datasets
- Dataset quality assessment

## Output

- Structured modeling-ready dataset

---

# D2 — System Identification Framework

## Objective

Develop behavioral/process model from acquired data.

## Sub-Tasks

- Define modeling strategy
    - physics-based
    - ML-based
    - hybrid modeling
- Select SI tools/frameworks
    - MATLAB SI toolbox
    - PyTorch/TensorFlow
- Define model inputs/outputs
- Implement baseline models
- Train and calibrate models
- Hyperparameter tuning
- Compare modeling approaches
- Build reusable SI workflow
- Model version management

## Output

- Identified behavioral model

---

# D3 — Offline Model Validation Framework (Validation Loop 1)

## Objective

Validate identified model before deployment/use.

## Sub-Tasks

- Train/test split definition
- Define validation KPIs
- Compute prediction accuracy metrics
- Residual/error analysis
- Sensitivity testing

- Stability checks
- Model confidence estimation
- Validation reporting
- Define model acceptance criteria

## Output

- Validated behavioral model
- Validation metrics/report

---

# D4 — Behavioral Digital Twin Framework

## Objective

Use identified model for virtual process simulation.

## Sub-Tasks

- Define DT architecture
- Integrate identified model into DT
- Build process simulation workflow
- Define virtual process inputs
- Develop prediction pipeline
- Implement scenario execution framework
- Simulate process behavior
- Generate predicted outputs
- KPI prediction framework
- Define DT interfaces/APIs

## Output

- Behavioral digital twin environment
- Predicted process outputs

---

# D5 — Process Intelligence, Simulation & Optimization Framework

## Objective

Provide framework for simulation, process analysis, and optimization studies.

## Sub-Tasks

- Develop what-if simulation capability
- Parameter variation studies
- Sensitivity analysis workflow
- KPI correlation framework
- Root-cause analysis workflow
- Bottleneck analysis framework
- Define optimization objectives
- Define optimization constraints
- Implement optimization workflow
- Parameter recommendation logic
- Multi-objective optimization studies
- Process insight generation
- Reporting/dashboard interfaces

## Output

- Simulation & optimization framework
- KPI insights
- Optimization recommendations

---

# D6 — Virtual Commissioning Integration

## Objective

Validate process and recommendations virtually.

## Sub-Tasks

- VC environment integration
- Process feasibility checks
- Constraint validation
- Virtual process execution
- Collision/reachability verification
- Validate recommended parameter sets
- Generate VC validation reports
- Integration with DT outputs

## Output

- Virtual validation results

---

# D7 — WP5.1 Visualization Interface

## Objective

Send predictive intelligence outputs to WP5.1.

## Sub-Tasks

- Define interface/API with WP5.1
- Define prediction output schema
- Define KPI visualization data structure
- Export predicted outputs
- Export model health indicators
- Export optimization insights
- Alarm/alert generation support
- Dashboard integration support
- Traceability metadata support

## Output

- Visualization-ready predictive outputs

---

# D8 — Online Validation & Continuous Learning Framework

## Objective

Continuously improve model using runtime data.

## Sub-Tasks

- Predicted vs actual comparison workflow
- Runtime error computation
- Drift detection framework
- Model performance monitoring
- Retraining trigger logic
- Incremental learning workflow
- Dataset update mechanism
- Re-validation workflow
- Continuous improvement pipeline

## Output

- Continuous learning workflow
- Model improvement indicators

---

# D9 — MES / Traceability Interface Support

## Objective

Provide integration interfaces for MES and traceability.

## Sub-Tasks

- Define MES integration interfaces
- Define optimization result export format
- Store process metadata

- Store model metadata
- Traceability record management
- Process history linkage
- Reporting interface support

## Output

- MES/traceability integration support

---

# Very Useful Management Summary

| Deliverable | Main Focus |
| --- | --- |
| D1 | Data preparation |
| D2 | Model development |
| D3 | Model validation |
| D4 | Digital twin simulation |
| D5 | Analysis & optimization framework |
| D6 | Virtual validation |
| D7 | Monitoring integration |
| D8 | Continuous learning |
| D9 | MES & traceability support |

These are the approximate dates and can. You give the subtaks date aligning with these

# D1 — Modeling Data Preparation (16-Jun-2026 → 21-Aug-2026)

## D1.1 Sample mTorres Data Collection & KPI Identification

### Subtasks

- Identify required process signals and KPIs
- Define data schema and tags
- Collect historical/sample machine data
- Validate signal availability
- Create initial dataset structure

### Timeline

16-Jun-2026 → 10-Jul-2026

---

## D1.2 Data Quality & Preprocessing

### Subtasks

- Timestamp synchronization
- Missing data handling
- Noise filtering
- Outlier detection
- Data normalization
- Feature extraction

### Timeline

11-Jul-2026 → 05-Aug-2026

---

## D1.3 Modeling Dataset Finalization

### Subtasks

- Generate train/validation/test datasets
- Dataset labeling
- Final data validation

- Curated dataset release

### Timeline

06-Aug-2026 → 21-Aug-2026

---

# D2 — Modeling Data Processing from WP5.1 (1-Oct-2026 → 22-Jan-2027)

## D2.1 WP5.1 Interface Definition

### Subtasks

- Define data exchange interface
- Define ingestion APIs/connectors
- Define data formats/schema

### Timeline

01-Oct-2026 → 25-Oct-2026

---

## D2.2 Data Ingestion Pipeline Development

### Subtasks

- Build ingestion workflow
- Implement synchronization logic
- Implement preprocessing pipeline
- Logging/error handling

### Timeline

26-Oct-2026 → 10-Dec-2026

---

## D2.3 Dataset Structuring & Validation

### Subtasks

- Dataset quality checks
- Curated dataset generation
- Validation against WP5.1 signals
- Documentation

### Timeline

11-Dec-2026 → 22-Jan-2027

---

# D3 — Defect / Design Model Development (1-Apr-2027 → 18-Jun-2027)

## D3.1 KPI / Process Correlation Study

### Subtasks

- KPI interaction analysis
- Process variable mapping
- Initial process understanding
- Correlation studies

### Timeline

01-Apr-2027 → 22-Apr-2027

---

## D3.2 Defect / Process Mapping

### Subtasks

- Defect-to-process mapping
- Label refinement
- Root-cause interaction study
- Sensitivity identification

### Timeline

23-Apr-2027 → 18-May-2027

---

## D3.3 Feature Engineering & Dataset Refinement

### Subtasks

- Feature selection
- Derived feature generation
- Dataset balancing
- Curated training dataset creation

### Timeline

19-May-2027 → 18-Jun-2027

---

# D4 — System Identification with Simulated Data (3-Aug-2026 → 4-Dec-2026)

## D4.1 SI Problem Definition

### Subtasks

- Define model inputs/outputs
- Define target process behavior
- Select SI methodology

### Timeline

03-Aug-2026 → 28-Aug-2026

---

## D4.2 SI Framework Setup

### Subtasks

- MATLAB/Python SI environment setup
- Define reusable workflows
- Data connectors setup

### Timeline

29-Aug-2026 → 25-Sep-2026

---

# D4.3 Initial Behavioral Model Development

### Subtasks

- Baseline model creation
- Parameter estimation
- Training/calibration
- Initial prediction validation

### Timeline

26-Sep-2026 → 30-Oct-2026

---

# D4.4 Offline Validation Loop 1

### Subtasks

- Train/test validation
- Residual analysis
- Error metrics computation
- Model tuning

### Timeline

31-Oct-2026 → 20-Nov-2026

---

# D4.5 Final Behavioral Model Release

### Subtasks

- Documentation
- Final validation
- Export reusable model
- Model readiness review

### Timeline

21-Nov-2026 → 04-Dec-2026

---

# D5 — System Identification Improvement with 5.1 Data (19-Jun-2027 → 3-Sep-2027)

## D5.1 Real Data Integration

### Subtasks

- Integrate WP5.1 datasets
- Align real signals with SI workflow
- Dataset synchronization

### Timeline

19-Jun-2027 → 10-Jul-2027

---

## D5.2 Model Retraining

### Subtasks

- Retrain behavioral model
- Compare simulated vs real behavior
- Improve robustness

### Timeline

11-Jul-2027 → 05-Aug-2027

---

## D5.3 Validation Loop 2 (Online Validation)

### Subtasks

- Predicted vs actual comparison
- Runtime error monitoring
- Drift analysis
- Model improvement workflow

### Timeline

06-Aug-2027 → 20-Aug-2027

## D5.4 Final Improved Model

### Subtasks

- Final validation
- Performance reporting
- Release updated behavioral model

### Timeline

21-Aug-2027 → 03-Sep-2027

---

# D6 — Analysis Framework & Digital Twin (1-Oct-2026 → 19-Mar-2027)

## D6.1 Behavioral Digital Twin Integration

### Subtasks

- Integrate identified model into DT
- Define DT execution pipeline
- Define simulation interfaces

### Timeline

01-Oct-2026 → 10-Nov-2026

---

## D6.2 Scenario Simulation Framework

### Subtasks

- What-if simulation capability
- Parameter variation studies
- Scenario execution logic

### Timeline

11-Nov-2026 → 20-Dec-2026

---

# D6.3 KPI Correlation & Insight Framework

### Subtasks

- KPI correlation workflow
- Sensitivity analysis
- Root-cause analysis framework

### Timeline

21-Dec-2026 → 31-Jan-2027

---

# D6.4 Optimization Framework

### Subtasks

- Define optimization objectives
- Constraint definition
- Recommendation logic
- Multi-objective optimization studies

### Timeline

01-Feb-2027 → 28-Feb-2027

---

# D6.5 WP5.1 Visualization Integration

### Subtasks

- Predicted output interface
- Model health visualization
- KPI insight export
- Dashboard integration support

### Timeline

01-Mar-2027 → 19-Mar-2027

Actually from 5.3 we don't have to do any of these Timestamp synchronization Missing data handling Noise filtering Outlier detection Data normalization Feature extraction This will be done by 5.1 Once we ingest data , we need to prepare the data for system identification purposes

Then your scope becomes much cleaner and more accurate architecturally.

Those activities:

- timestamp synchronization
- filtering
- noise handling
- normalization
- preprocessing

belong to:

# ✅ WP5.1 — Data Processing / Monitoring Layer

NOT WP5.3.

WP5.3 should assume:

“cleaned and curated process data is available”

Then WP5.3 focuses on:

# ✅ Modeling-oriented preparation

for:

- SI
- DT
- optimization

---

# Corrected WP5.3 Scope

Instead of:

Data preprocessing

Use:

Modeling dataset preparation

---

# Corrected D1

# D1 — Modeling Data Preparation

## Objective

Prepare WP5.1 curated data for System Identification workflows.

---

# Correct Subtasks

## D1.1 Define SI Modeling Variables

- Define model inputs/outputs
- Define KPIs for SI
- Define target process behavior
- Define process states/labels

---

## D1.2 Modeling Dataset Structuring

- Select relevant process signals
- Windowing/sequence preparation
- Train/validation/test split
- Dataset formatting for MATLAB/Python SI frameworks
- Scenario grouping

---

## D1.3 Feature & Label Preparation for SI

- Derived feature preparation for modeling
- Defect/event labeling for learning
- Behavioral state classification
- Dataset balancing

---

## D1.4 Dataset Validation for SI Readiness

- Validate completeness for modeling
- Validate SI compatibility
- Validate model coverage
- Curated modeling dataset release

---

# MUCH BETTER ARCHITECTURE

| WP5.1 | WP5.3 |
| --- | --- |
| Data acquisition | System identification |
| Signal synchronization | SI dataset preparation |
| Filtering | Behavioral modeling |
| Noise removal | Digital twin |
| Preprocessing | Simulation |
| KPI calculation | Optimization |

This separation is architecturally correct.

---

# Important wording change

Instead of:

Data preprocessing

Use:

Model-oriented dataset preparation

or

Dataset preparation for behavioral modeling

---

# Updated realistic subtasks

## What WP5.3 should actually do

### Inputs from WP5.1

Clean synchronized process dataset

### WP5.3 tasks

Select → structure → label → prepare
for SI and DT

NOT:

Acquire → filter → clean

---

# Best concise wording for roadmap

## Deliverable

Modeling dataset preparation

## Output

SI-ready behavioral modeling dataset

This is probably the most accurate phrasing for your scope.

TASKS

Your structure is actually quite strong already. The overall sequence is technically logical and much better than a typical high-level R&T roadmap.

You already have:

- architecture phase
- sample-data phase
- SI phase
- DT phase
- optimization phase
- real-data refinement phase
- VC phase
- referencing phase
- MES phase

which is a very good decomposition.

But yes — there are:

1. some repetitions
2. some misplaced tasks
3. some missing engineering tasks
4. some naming inconsistencies
5. some deliverable overlaps

Below is a detailed review.

---

# 1. Biggest Issue — D Number Reuse / Conflicts

You reused:

- D4
- D5

- D6
- D7
- D8

for multiple independent domains.

This will become very confusing later during:

- reviews
- audits
- milestone tracking
- presentations
- Jira linkage

---

# Recommendation

Keep separate numbering domains.

Example:

| Domain | Suggested Prefix |
| --- | --- |
| HDT/SI | HDT-Dx |
| Optimization | OPT-Dx |
| VC | VC-Dx |
| Referencing | REF-Dx |
| MES | MES-Dx |

OR simpler:

| Current | Better |
| --- | --- |
| D4 Ver1 SI | D4 |
| D5 Ver2 SI | D5 |
| D6 Ver1 Analysis | D6 |
| D7 Ver2 Analysis | D7 |
| D8 Visualization | D8 |
| D6 Optimization | D9 |
| D7 Real-time prediction | D10 |
| D8 VC | D11 |

Otherwise people will get lost.

This is the MOST important cleanup.

---

# 2. D2 is GOOD overall

D2 structure is actually very solid.

You correctly separated:

- signal understanding
- SI variable planning
- dataset structuring
- feature/label prep
- validation

Very good.

---

# 3. D2.5 — Slightly Too Detailed

This section is technically excellent, but roadmap/task sheets should not contain too much theory.

Example:

Power-to-Force ratios to capture actual physical load...

Too deep for milestone planning sheet.

Move those explanations into:

- technical note
- architecture doc
- methodology doc

Keep roadmap concise.

---

# Recommendation

Shorten to:

## D2.5 Feature & Label Preparation for SI

- Derived feature generation
- Defect/event labeling
- Behavioral state classification

- Dataset balancing
- Time-delay feature alignment

Much cleaner.

---

# 4. D3 — Structure Needs Cleanup

Currently D3 mixes:

- ingestion architecture
- defect mapping
- final dataset preparation

These are different maturity levels.

---

# Recommended D3 Structure

# D3 — Data Ingestion from WP5.1 — Design & Development

## D3.1 Interface Contract Definition

- schema
- metadata
- naming
- APIs
- signal definitions

---

## D3.2 Ingestion Pipeline Development

- ingestion framework
- parser

- storage
- versioning
- simulator adapter

---

### D3.3 WP5.1 Integration Validation

- dry-run integration
- schema validation
- stability validation

---

Then move:

- defect mapping
- modeling-ready dataset preparation

to D4.

This is cleaner architecturally.

---

# 5. D4 (Actual laying data) — Missing Tasks

D4 currently almost empty.

You need the tasks we discussed.

---

# Recommended D4 Structure

# D4 — Actual Laying Data Integration & Modeling Preparation

## D4.1 Actual Data Reception

- receive actual laying data
- validate completeness
- dataset registration

---

## D4.2 SI Variable Refinement

- validate SI inputs/outputs
- refine KPIs
- refine process states
- refine prediction targets

---

## D4.3 Modeling Dataset Preparation

- train/validation/test preparation
- sequence generation
- scenario grouping
- DT datasets

---

## D4.4 Defect-to-Process Mapping

- align defects to process windows
- initial root-cause mapping

---

## D4.5 Dataset Validation

- SI compatibility
- DT compatibility
- prediction coverage validation

This is one of the biggest missing sections.

---

# 6. D4 Ver1 SI — Missing Important Engineering Tasks

You have:

- develop SI
- system mapping
- algorithms

Good.

But missing:

- baseline model comparison
- model calibration
- residual analysis
- offline validation loop

---

# Add

## D4.x Offline Validation Loop 1

- train/test validation
- residual analysis
- error metrics
- model tuning

Very important technically.

---

# 7. D5 Ver2 SI Improvement — GOOD

Actually good structure.

No major issues.

Maybe add:

- drift analysis
- online validation

---

# Add

## D5.x Runtime Model Monitoring

- prediction vs actual
- drift monitoring
- retraining trigger logic

Because this becomes your:

Validation Loop 2

---

# 8. D6/D7 Analysis + DT — Some Repetition

You currently split:

- Ver1 with sample data
- Ver2 with actual data

This is okay.

But:

- KPI correlation
- sensitivity
- RCA

are repeated.

---

# Better Structure

### D6 Ver1

Develop framework with sample data

### D7 Ver2

Validate/refine framework using actual data

NOT redevelop.

---

# Rename D7 Tasks

Instead of:

Framework for KPI correlation analysis

Use:

Validation and refinement of KPI correlation framework using actual laying data

This avoids repetition.

---

# 9. Optimization Section — Missing Constraints Definition

Optimization without constraints is dangerous.

Add:

## Optimization Constraint Framework

- process limits
- machine limits
- safety constraints
- feasible operating windows

Very important industrially.

---

# 10. Real-time Prediction Section — VERY GOOD

Actually strong.

Maybe rename:

Predictive indicators

to:

Early warning indicator framework

More professional.

---

# 11. VC Section — Missing Digital Twin Connection

Currently VC appears isolated.

Need explicit linkage:

## Add

Integration of behavioral DT outputs with VC validation

Otherwise reviewers may ask:

How VC connects to HDT?

---

# 12. Referencing Section — VERY GOOD

Actually one of the cleanest sections.

Only missing:

### Add

Reference frame transformation validation

between:

- localization
- NC correction

---

# 13. MES Section — Too High Level

Need:

- traceability
- storage
- metadata
- optimization history

---

# Add

### MES Data Mapping

- optimization result storage
- DT metadata storage
- traceability mapping

---

# 14. BIGGEST Missing Deliverable

# Missing:

### Continuous Learning / Runtime Validation

You discussed it extensively but roadmap barely reflects it.

Need explicit deliverable.

---

# Recommended Addition

# D10 — Runtime Validation & Continuous Learning

### Tasks

- predicted vs actual comparison
- model health monitoring
- drift analysis
- retraining trigger logic
- runtime model validation
- continuous model improvement

This is VERY important because it differentiates:

- static SI
  vs
- intelligent HDT framework

---

# Final Overall Assessment

| Area | Status |
| --- | --- |

| | |
|---|---|
| Architecture flow | Very good |
| Technical depth | Strong |
| Deliverable sequencing | Good |
| Numbering consistency | Needs cleanup |
| Runtime validation | Missing |
| D4 actual data phase | Incomplete |
| VC ↔ DT linkage | Missing |
| Repetition handling | Needs refinement |
| Industrial realism | Strong |

Overall:

~80–85% mature roadmap already

Very solid for an R&T planning sheet.

can you give the list of tasks with little more information not just as bullet points and give the updated list

Below is the updated and refined task structure for WP5.3 with:

- repetitions removed
- architecture aligned

- clearer task descriptions
- proper separation between WP5.1 and WP5.3 responsibilities
- more engineering-oriented wording

This should be much more suitable for:

- roadmap reviews
- management discussions
- architecture reviews
- Jira/Epic decomposition
- TRL planning

Based on your uploaded planning sheet.

---

# D0 — Requirement Gathering

## D0.1 Gather HDT, VC and Referencing Requirements

### Tasks

- Conduct workshops with OMTC and stakeholders to understand expected Hybrid Digital Twin (HDT), Virtual Commissioning (VC), and referencing capabilities.
- Identify expected process monitoring, prediction, optimization, and correction use cases.
- Collect constraints related to machine architecture, process flow, sensors, and deployment environment.
- Identify expected KPIs, defect indicators, and operational pain points.

### Output

- Initial requirement specification
- Use-case definition
- Stakeholder alignment report

---

## D0.2 Validation of Requirements with OMTC

### Tasks

- Review gathered requirements with OMTC and domain experts.

- Validate technical feasibility and scope boundaries.
- Finalize project priorities and deliverable expectations.
- Align terminology, KPIs, and operational definitions across teams.

### Output

- Approved requirement baseline

---

# D1 — Hybrid Digital Twin Architecture & ICD

## D1.1 Overall System Architecture Definition

### Tasks

- Define the end-to-end Hybrid Digital Twin workflow:

  data → behavioral model → simulation → optimization → correction → feedback

- Define interactions between WP5.1, WP5.2, MES, VC, referencing, and optimization modules.
- Define high-level system boundaries and responsibilities.
- Define runtime and offline execution flows.

### Output

- High-level HDT architecture

---

## D1.2 Functional Block Design

### Tasks

- Define major functional modules:
  - System Identification
  - Behavioral Digital Twin
  - Simulation Framework

    - Optimization Framework
    - Referencing
    - NC Correction
    - MES Integration
    - Virtual Commissioning
- Define responsibilities and interfaces of each block.
- Define expected inputs/outputs for each module.

### Output

- Functional architecture document

---

## D1.3 Data Flow & Interface Definition (ICD)

### Tasks

- Define how data moves between:
    - WP5.1
    - SI models
    - Digital Twin
    - MES
    - VC
- Define interface contracts and data exchange mechanisms.
- Define metadata structure and communication patterns.
- Define traceability requirements for exchanged data.

### Output

- Interface Control Document (ICD)

---

## D1.4 Referencing Integration Architecture

### Tasks

- Define how referencing/localization outputs update NC correction and DT environments.
- Define coordinate transformation workflow.
- Define integration with external metrology systems.

### Output

- Referencing integration workflow

---

# D1.5 NC Correction Strategy Definition

### Tasks

- Define static NC correction methodology:

  OLP → New Reference Frame → Corrected NC

- Define transformation logic for TCP positions and orientations.
- Define corrected NC output interfaces.

### Output

- NC correction strategy document

---

# D1.6 MES Integration Architecture

### Tasks

- Define MES interaction points.
- Define process traceability workflow.
- Define optimization and DT metadata storage strategy.

### Output

- MES integration architecture

---

# D1.7 Virtual Commissioning Integration

### Tasks

- Define how VC validates:
  - behavioral DT outputs
  - optimized parameters
  - NC corrections

- Define simulation-to-deployment validation workflow.

### Output

- VC integration workflow

---

# D2 — Modeling Data Preparation (Sample mTorres Data)

## D2.1 Identify Required Process Signals & KPIs

### Tasks

- Identify process variables relevant for modeling.
- Define process KPIs influencing quality and defects.
- Identify machine, sensor, and contextual signals.
- Identify relationships between process behavior and defects.

### Output

- Initial process signal list
- KPI definition document

---

## D2.2 Collect Historical/Sample Machine Data

### Tasks

- Gather available mTorres sample datasets.
- Organize process, machine, and sensor datasets.
- Review process variability and available defect information.
- Build initial data repository for SI activities.

### Output

- Initial sample dataset repository

---

# D2.3 Define SI Modeling Variables (Planning Phase)

### Tasks

- Define model inputs and prediction outputs.
- Define target process behaviors to be represented by the SI model.
- Define process operating states:
    - Laydown
    - Travel
    - Cut/Add
    - Retract
- Define behavioral labels and state transitions.
- Validate KPIs and prediction targets with stakeholders.

### Output

- SI variable mapping
- Behavioral state definition

---

# D2.4 Modeling Dataset Structuring

### Tasks

- Select signals relevant for behavioral modeling.
- Prepare time-sequence/window-based datasets.
- Organize datasets into:
    - train
    - validation
    - test
- Group datasets by manufacturing scenarios and operating conditions.
- Format datasets for MATLAB/Python SI frameworks.

### Output

- Structured SI-ready datasets

---

## D2.5 Feature & Label Preparation for SI

### Tasks

- Generate derived features useful for modeling.
- Align defect and event labels with process timelines.
- Prepare behavioral state classifications.
- Balance datasets to improve model robustness.
- Align delayed variables and time-dependent process effects.

### Output

- Feature-engineered labeled datasets

---

## D2.6 Dataset Validation for SI Readiness

### Tasks

- Validate dataset completeness and coverage.
- Validate compatibility with SI workflows.
- Validate coverage across different process conditions.
- Release curated dataset for initial SI activities.

### Output

- Validated SI-ready dataset

---

# D3 — Data Ingestion from WP5.1 — Design & Development

## D3.1 Interface Contract Definition

### Tasks

- Define schema for data exchange between WP5.1 and WP5.3.
- Define signal naming conventions, metadata, units, and timestamps.

- Define ingestion frequency and synchronization expectations.
- Define storage and traceability structure.

### Output

- WP5.1 ↔ WP5.3 data interface specification

---

## D3.2 Data Ingestion Pipeline Development

### Tasks

- Develop ingestion framework/API for WP5.1 datasets.
- Develop dataset parsers and validation logic.
- Implement dataset registration and versioning.
- Build storage structure for SI and DT workflows.
- Develop sample-data simulator adapter for pipeline testing.

### Output

- Reusable ingestion pipeline

---

## D3.3 WP5.1 Integration Readiness Validation

### Tasks

- Validate ingestion workflow using mock/sample datasets.
- Validate schema compatibility and extensibility.
- Perform dry-run integration testing.
- Validate ingestion stability and traceability.

### Output

- Validated ingestion architecture

---

# D4 — Actual Laying Data Integration & Modeling Preparation

## D4.1 Actual Laying Data Reception

### Tasks

- Receive curated laying datasets from WP5.1.
- Import datasets through developed ingestion pipeline.
- Validate dataset completeness and metadata availability.
- Register and organize incoming datasets.

### Output

- Integrated actual laying datasets

---

## D4.2 SI Variable Refinement using Actual Laying Data

### Tasks

- Validate and refine SI inputs/outputs using real production data.
- Refine process states and behavioral labels.
- Refine KPI definitions and prediction targets.
- Validate process-state segmentation against actual process behavior.

### Output

- Refined behavioral modeling variables

---

## D4.3 Modeling Dataset Preparation

### Tasks

- Prepare SI-ready datasets using actual laying data.
- Generate scenario-based and sequence-based datasets.
- Prepare datasets for DT simulation and optimization workflows.

- Organize datasets by process conditions and defect classes.

### Output

- SI/DT-ready structured datasets

---

## D4.4 Defect-to-Process Mapping

### Tasks

- Link defects to corresponding process windows and trajectories.
- Map process conditions to defect occurrence.
- Create initial root-cause relationship mappings.

### Output

- Defect-process correlation dataset

---

## D4.5 Dataset Validation for SI & DT Usage

### Tasks

- Validate dataset compatibility with SI and DT workflows.
- Validate prediction target availability.
- Validate scenario coverage and operational diversity.

### Output

- Validated production-ready modeling dataset

---

(continued in your later deliverables: SI, DT, Optimization, VC, Runtime Validation, MES, Referencing)

can you give the list of tasks again with apprpximate days or months required to complete it

# D0 — Requirement Gathering

## Total Duration: ~2.5 Months

| Task | Approx Duration |
|---|---|
| D0.1 Gather HDT, VC and referencing requirements | 6–7 weeks |
| D0.2 Requirement validation with OMTC | 2 weeks |

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# D1 — Hybrid Digital Twin Architecture & ICD

## Total Duration: ~5 Months

| Task | Approx Duration |
|---|---|
| D1.1 Overall system architecture definition | 3 weeks |
| D1.2 Functional block design | 6–8 weeks |
| D1.3 Data flow & interface definition (ICD) | 8–10 weeks |
| D1.4 Referencing integration architecture | 2–3 weeks |
| D1.5 NC correction strategy definition | 2–3 weeks |

| | |
|---|---|
| D1.6 MES integration architecture | 2 weeks |
| D1.7 Virtual commissioning integration definition | 2–3 weeks |

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# D2 — Modeling Data Preparation (Sample mTorres Data)

## Total Duration: ~2 Months

| Task | Approx Duration |
|---|---|
| D2.1 Identify required process signals & KPIs | 2 weeks |
| D2.2 Collect historical/sample machine data | 2 weeks |
| D2.3 Define SI modeling variables & behavioral states | 2–3 weeks |
| D2.4 Modeling dataset structuring | 2 weeks |
| D2.5 Feature & label preparation for SI | 3–4 weeks |
| D2.6 Dataset validation for SI readiness | 2 weeks |

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# D3 — Data Ingestion from WP5.1 — Design & Development

## Total Duration: ~2.5 Months

| Task | Approx Duration |
|---|---|
| D3.1 Interface contract definition | 3 weeks |
| D3.2 Data ingestion pipeline development | 6–7 weeks |
| D3.3 WP5.1 integration readiness validation | 2–3 weeks |

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# D4 — Actual Laying Data Integration & Modeling Preparation

## Total Duration: ~3.5 Months

| Task | Approx Duration |
|---|---|
| D4.1 Actual laying data reception & integration | 2 weeks |
| D4.2 SI variable refinement using actual laying data | 3 weeks |

| | |
|---|---|
| D4.3 Modeling dataset preparation | 4 weeks |
| D4.4 Defect-to-process mapping | 3 weeks |
| D4.5 Dataset validation for SI & DT usage | 2 weeks |

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# D5 — Ver1 System Identification (Sample Data / POC Phase)

## Total Duration: ~4 Months

| Task | Approx Duration |
|---|---|
| D5.1 System I/O mapping & KPI categorization | 2 weeks |
| D5.2 Synthetic proxy plant / MATLAB-Python environment setup | 3 weeks |
| D5.3 Implement SI algorithms | 5–6 weeks |
| D5.4 Initial model training & calibration | 3 weeks |
| D5.5 Offline validation loop 1 (train/test validation) | 3 weeks |
| D5.6 Residual analysis & model refinement | 2 weeks |

| | |
|---|---|
| D5.7 Initial behavioral model release | 1 week |

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# D6 — Ver2 System Identification Improvement with WP5.1 Data

## Total Duration: ~2.5 Months

| Task | Approx Duration |
|---|---|
| D6.1 Input-output mapping refinement | 2 weeks |
| D6.2 Model structure refinement | 2 weeks |
| D6.3 Model retraining using actual laying data | 3 weeks |
| D6.4 Model validation & residual analysis | 2 weeks |
| D6.5 Runtime drift analysis & monitoring | 2 weeks |

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# D7 — Ver1 Analysis Framework & Digital Twin (Sample Data)

## Total Duration: ~5.5 Months

| Task | Approx Duration |
|---|---|
| D7.1 Behavioral DT integration | 4 weeks |
| D7.2 Scenario simulation framework | 4 weeks |
| D7.3 Root cause analysis framework | 3 weeks |
| D7.4 KPI correlation framework | 3 weeks |
| D7.5 Sensitivity analysis framework | 3 weeks |
| D7.6 Predicted output visualization framework | 2 weeks |
| D7.7 DT validation using sample scenarios | 2 weeks |

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# D8 — Ver2 Analysis Framework & DT using Actual WP5.1 Data

## Total Duration: ~1.5–2 Months

| Task | Approx Duration |
|---|---|

| | |
|---|---|
| D8.1 DT refinement using actual laying data | 2 weeks |
| D8.2 Validation of KPI correlation framework | 2 weeks |
| D8.3 Validation of root cause framework | 2 weeks |
| D8.4 Validation of sensitivity analysis | 1–2 weeks |
| D8.5 Validation of predicted outputs | 1–2 weeks |

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# D9 — Visualization Integration with WP5.1

## Total Duration: ~3.5 Months

| Task | Approx Duration |
|---|---|
| D9.1 Define visualization interface | 2 weeks |
| D9.2 Predicted output integration with WP5.1 | 4 weeks |
| D9.3 Model health & KPI visualization | 3 weeks |
| D9.4 Dashboard integration support | 3 weeks |

| | |
|---|---|
| D9.5 Validation with WP5.1 | 2 weeks |

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# D10 — Optimization Framework (Sample Data)

## Total Duration: ~4.5 Months

| Task | Approx Duration |
|---|---|
| D10.1 Optimization objective definition | 2 weeks |
| D10.2 Constraint framework definition | 3 weeks |
| D10.3 Optimization engine development | 5 weeks |
| D10.4 Parameter recommendation logic | 3 weeks |
| D10.5 Optimization studies using sample data | 3 weeks |
| D10.6 Validation of optimization workflow | 2 weeks |

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# D11 — Optimization Framework with Actual Laying Data

## Total Duration: ~2 Months

| Task | Approx Duration |
|---|---|
| D11.1 Safe operating window definition | 2 weeks |
| D11.2 Optimization refinement using real data | 3 weeks |
| D11.3 Validation of optimization recommendations | 2 weeks |
| D11.4 Final optimization framework release | 1 week |

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# D12 — Runtime Prediction & Continuous Learning

## Total Duration: ~3 Months

| Task | Approx Duration |
|---|---|
| D12.1 Real-time defect prediction framework | 4 weeks |

| | |
|---|---|
| D12.2 Deviation detection framework | 3 weeks |
| D12.3 Early warning indicator framework | 3 weeks |
| D12.4 Runtime validation loop (Prediction vs Actual) | 3 weeks |

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# D13 — Virtual Commissioning

## Total Duration: ~15–16 Months

| Task | Approx Duration |
|---|---|
| D13.1 VC architecture definition | 4 months |
| D13.2 Simulation model creation | 3–4 months |
| D13.3 Integration of simulation models | 3 months |
| D13.4 VC testing & validation | 2 months |
| D13.5 Shop floor integration support | 5 months |
| D13.6 HDT + VC integration testing | 3 months |

| | |
|---|---|
| D13.7 End-to-end integration with DAQ | 3 months |

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# D14 — Zero-Time Roto-Translation / Referencing

## Total Duration: ~10 Months

| Task | Approx Duration |
|---|---|
| D14.1 CAD feasibility & line-of-sight study | 2–3 weeks |
| D14.2 Referencing system architecture definition | 2 months |
| D14.3 Electrical design | 3–4 weeks |
| D14.4 Procurement & launch readiness | 1–2 months |
| D14.5 Mechanical structure & target support design | 1–2 months |
| D14.6 Referencing software development | 3 months |
| D14.7 Integration & POC | 1–1.5 months |
| D14.8 Testing & validation | 1 month |

| | |
|---|---|
| D14.9 Industrialization documentation | 2 weeks |
| D14.10 Referencing output interface to NC correction | 2 weeks |

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# D15 — MES Integration & Feedback Loop

## Total Duration: ~3–4 Months

| Task | Approx Duration |
|---|---|
| D15.1 MES integration architecture | 3 weeks |
| D15.2 MES integration implementation | 6 weeks |
| D15.3 Traceability & metadata integration | 2 weeks |
| D15.4 Workflow testing & operator validation | 3 weeks |
| D15.5 Feedback loop validation | 2 weeks |